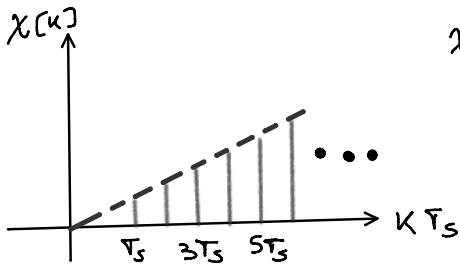


Repaso transformada Z:

$$X(z) = \sum_{k=0}^{+\infty} x[k] z^{-k}$$

Rampa:

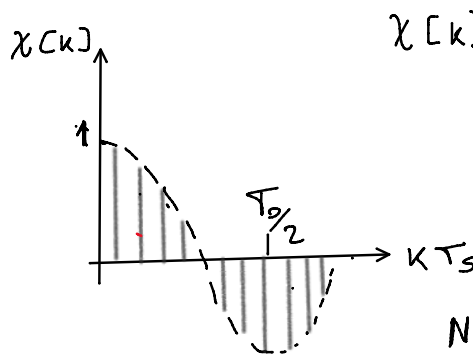
$$x[k] = k u[k] \quad k = 0, 1, 2, \dots$$

$$X(z) = \sum_{k=0}^{+\infty} k \cdot z^{-k} = 0 + \frac{1}{z} + \frac{1}{z^2} + \dots$$

Propiedad: $k x[k] \xrightarrow{Z} -z \cdot \frac{dX(z)}{dz} \quad \therefore X(z) = -z \frac{d\left(\sum_{k=0}^{+\infty} u[k] z^{-k}\right)}{dz}$

$$X(z) = -z \left(\frac{z}{z-1} \right)' = \frac{(z-1) - z}{(z-1)^2} (-z)$$

$$X(z) = \frac{z}{(z-1)^2}$$

Coseno:

$$x[k] = \cos\left(k \frac{2\pi}{T_0} T_s\right)$$

$$= \cos\left(2\pi \frac{T_s}{T_0} k\right)$$

$$= \cos\left(2\pi \frac{k}{N}\right)$$

$$N = \text{int}(T_0/T_s) \rightarrow \omega_N = 2\pi/N$$

$$x[k] = \cos(\omega_N k) = \frac{1}{2} \left[e^{j\omega_N k} + e^{-j\omega_N k} \right]$$

Propiedad: $e^{-jk} \cdot x[k] \xrightarrow{Z} X(z \cdot e^{+2})$

$$x[k] = \frac{1}{2} \left[e^{j\omega_N k} u[k] + e^{-j\omega_N k} u[k] \right]$$

$$\mathbb{Z}\{x[k]\} = \frac{1}{2} \left\{ \mathbb{Z}\{e^{j\omega_N k} \cdot u[k]\} + \mathbb{Z}\{e^{-j\omega_N k} \cdot u[k]\} \right\}$$

$$= \frac{1}{2} \left[U(z \cdot e^{-j\omega_N}) + U(z \cdot e^{j\omega_N}) \right]$$

$$\text{Donde: } U(z) = \mathbb{Z}\{u[k]\}$$

$$X(z) = \frac{1}{2} \left[\frac{1}{1 - (z^{-1} e^{j\omega_N})} + \frac{1}{1 - (z^{-1} e^{-j\omega_N})} \right]$$

Operando

$$X(z) = \frac{z^2 + z \cos(\omega_N)}{1 - 2z \cos(\omega_N) + z^2}$$

Transformada Z Inversa:

$$\text{Por Fracciones Simples: } \frac{X(z)}{z} = \frac{a_1}{z-1} + \frac{a_2}{z-e^{-j\omega_N}}$$

$$X(z) = \frac{(1-e^{-j\omega_N}) \cdot z}{(z-1)(z-e^{-j\omega_N})}$$

$$X(z)/z = \frac{K}{(z-p_1)(z-p_2)} \quad \begin{cases} K = 1-e^{-j\omega_N} \\ p_1 = 1 \\ p_2 = e^{-j\omega_N} \end{cases}$$

$$a_1 = \left[\frac{(\cancel{z-p_1}) K}{(\cancel{z-p_1})(z-p_2)} \right] \bigg|_{z=p_1} = \frac{K}{p_1-p_2} = \frac{1-e^{-j\omega_N}}{1-e^{-j\omega_N}} = 1$$

$$a_2 = \left[\frac{(\cancel{z-p_2}) K}{(z-\cancel{p_2})} \right] \bigg|_{z=p_2} = \frac{K}{p_2-p_1} = \frac{1-e^{-j\omega_N}}{-1+e^{-j\omega_N}} = -1$$

$$X(z) = z \left[\frac{1}{z-1} - \frac{1}{z-e^{-j\omega_N}} \right] = \frac{z}{z-1} - \frac{z}{z-e^{-j\omega_N}}$$

$$X(z) = z \left[\frac{1}{z-1} - \frac{1}{z-e^{-aT}} \right] = \frac{z}{z-1} - \frac{z}{z-e^{-aT}}$$

$$\stackrel{Z^{-1}}{\rightarrow} x[k] = u[k] - e^{-aT} u[k]$$

$$x[k] = u[k] \cdot (1 - e^{-aT})$$

Ecuaciones en Diferencia:

$$x[k+2] + 3x[k+1] + 2x[k] = 0 \quad x[0] = 0 \quad ; \quad x[1] = 1$$

$$Z \{ x[k+2] \} = z^2 \cdot X(z) - z \cdot x[0] - z x[1]$$

$$Z \{ x[k+1] \} = z \cdot X(z) - z \cdot x[0]$$

$$Z \{ x[k] \} = X(z)$$

$$z^2 X(z) - z^2 x[0] - z x[1] + 3z X(z) - 3z x[0] + 2X(z) = 0$$

Reemplazo $\rightarrow X(z) = \frac{z}{z^2 + 3z + 2} = \frac{z}{(z+1)(z+2)}$

$$X(z) = \frac{z}{z+1} - \frac{z}{z+2} \quad \stackrel{Z^{-1}}{\rightarrow} \quad x[k] = (-1)^k + (-2)^k$$